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B.Tech. Degree VI Semester Examination April 2015**ME 1605 CAD/CAM**
(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Mention the role of FEM in engineering design.
(b) What is Detroit model of automation?
(c) What are the advantages of CNC?
(d) Write a note on APT.
(e) List out the dynamic and static errors in CNC machine.
(f) What are the testing methods of NC tools?
(g) Describe FMS.
(h) What are the applications of robots?

PART B

(4 × 15 = 60)

- II. (a) Explain wire frame and solid modeling.
(b) Describe data exchange between CAD and CAM.
- OR**
- III. What is line balancing? Describe line balancing methods with suitable examples.
- IV. (a) Classify different CNC tools, with neat sketches.
(b) What are the control systems in CNC machines?
- OR**
- V. What is CAPP? Explain the various methods in CAPP.
- VI. What are the special design features of CNC machines?
- OR**
- VII. Explain (i) ATC (ii) APC.
- VIII. (a) Explain the classification of robots, with neat sketches.
(b) Write a note on robotic programming.
- OR**
- IX. (a) Describe CIM.
(b) Explain AI and expert systems.

